

Regularization

Regularization : (Part 1)

⇒ What is Regularization? It's a technique used to reduce overfitting by adding a penalty for overly large model coefficients/parameters to the loss function.

⇒ Why do we need it? Suppose Polynomial Regression has a very high degree:

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_{15} x^{15}$$

It may create a highly complicated curve that passes almost perfectly through the training points, i.e., low bias + high variance.

Regularization discourages such extreme parameter values, making the model less sensitive to the training data.

∴ main goal is:

Reduce Variance / Overfitting

⇒ Types of Regularization:

① Ridge Regression — L2 Regularization

② Lasso Regression — L1 Regularization

③ Elastic Net — Combⁿ of L1 and L2.

⇒ Ridge Regression: It is L.R. with an L2 regularization penalty that discourages large values of the model coefficients.

⇒ Why it's called L2? Bcz the penalty uses the squared values of the coefficients.

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda m^2$$

↑ Slope ↑ regularization strength

⇒ What's the whole idea behind Ridge? Ridge says to, "large coeff. are costly. Try to keep them smaller unless they are really necessary to reduce the prediction error".

So, Ridge tries to find a balance:

Fit the data well + keep coeff. small

⇒ Ridge primarily aims to reduce variance making sure bias-variance tradeoff is there.

Regularisation : (Part-2)

⇒ Mathematical Formulation:

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda m^2$$

$$\Rightarrow L = \sum (y_i - mx_i - b)^2 + \lambda m^2$$

$$\therefore \frac{\partial L}{\partial b} = 0 \Rightarrow b = \bar{y} - m\bar{x}$$

$$\therefore L = \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})^2 + \lambda m^2$$

$$\therefore \frac{\partial L}{\partial m} = 2 \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda m = 0$$

$$\Rightarrow \frac{\partial L}{\partial m} = \lambda m - \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + m \sum_{i=1}^n (x_i - \bar{x})^2 = 0$$

$$\Rightarrow m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

where, λ = hyperparameter

$$\therefore m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (\text{for LR})$$

∴ λ is helping to change the m .

⇒ For ω :

$$\omega = (X^T X)^{-1} X^T Y$$

Regularization : (Part 3)

⇒ Ridge Regression using Gradient descent:

We can write loss in vector form as:

$$L = (X\omega - Y)^T (X\omega - Y) + \lambda \|\omega\|^2$$

$$\Rightarrow L = (X\omega - Y)^T (X\omega - Y) + \lambda \omega^T \omega$$

$$\omega_{\text{new}} = \omega_{\text{old}} - \eta \frac{\Delta L}{\Delta \omega}$$

where,

$$\frac{\Delta L}{\Delta \omega} = \text{gradient}$$

$$= \begin{bmatrix} \partial L / \partial \omega_0 \\ \partial L / \partial \omega_1 \\ \vdots \\ \partial L / \partial \omega_n \end{bmatrix}$$

Now,

$$L = (X\omega - Y)^T (X\omega - Y) + \lambda \omega^T \omega$$

$$\Rightarrow L = (\omega^T X^T - Y^T)(X\omega - Y) + \lambda \omega^T \omega$$

$$\Rightarrow L = \frac{1}{2} (\omega^T X^T - Y^T)(X\omega - Y) + \frac{1}{2} \lambda \omega^T \omega$$

$$\Rightarrow L = \frac{1}{2} [\omega^T X^T X \omega - 2Y^T X \omega + Y^T Y] + \frac{1}{2} \lambda \omega^T \omega$$

$$\therefore \frac{\partial L}{\partial \omega} = \frac{\Delta L}{\Delta \omega} = X^T X \omega - X^T Y + \lambda \omega$$

$$\omega = \begin{bmatrix} \omega_0 & \omega_1 & \dots & \omega_n \\ 0 & 1 & \dots & 1 \end{bmatrix} \text{ starting}$$

epochs,

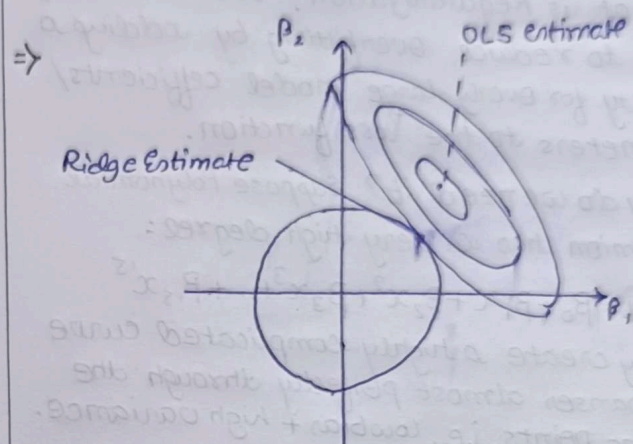
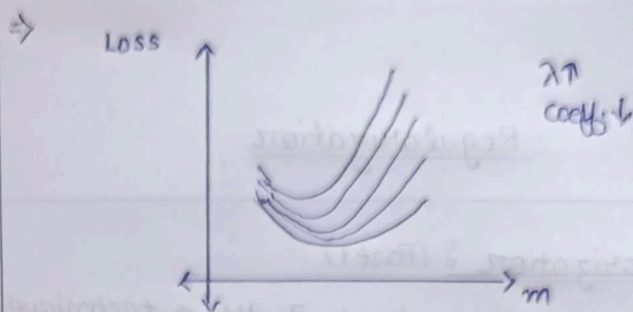
$$\omega = \omega - \eta \frac{\partial L}{\partial \omega}$$

Regularization: (Part 4)

$\Rightarrow$ How do the coefficients get affected? Ridge shrinks the coefficients towards zero. Ridge doesn't make coefficients exactly zero.

$\Rightarrow$ Ridge strongly discourages very large coefficients. Large coeffs. get disproportionately large penalties.

$\Rightarrow \lambda \uparrow \Rightarrow \text{Bias} \uparrow, \text{Variance} \downarrow$



Lasso Regression:

$\Rightarrow$ What is Lasso Regression? It's a regularised form of Linear Regression that adds an L1 penalty to the loss function, shrinking coefficients towards zero and potentially making some coefficients exactly zero.

$\Rightarrow$ LASSO stands for: Least Absolute Shrinkage and Selection Operator,

- Least $\rightarrow$ minimize a loss
- Absolute $\rightarrow$ uses $|\beta_j|$ or $|\omega|$
- Shrinkage $\rightarrow$ shrinks coeff. towards zero.
- Selection $\rightarrow$ can make coeff. exactly zero, thereby selecting features.

$\Rightarrow$ Lasso uses:

$$L = \text{MSE} + \lambda \|\omega\|$$

$$= \text{MSE} + \lambda [|\omega_1| + |\omega_2| + \dots + |\omega_n|]$$

⇒ Why Lasso is called L1 Regularization? Bcz, it uses the absolute value of the coefficients.

∴ Ridge : $\sum \omega^2 \rightarrow L2$

Lasso : $\sum \omega \rightarrow L1$

⇒ we use Lasso when we want regularization + automatic feature selection.

⇒

	Ridge	Lasso
Penalty	ω^2	ω
Effect	Shrinks coeff.	Shrinks coeff.
Can make coeff. exactly 0?	Generally No	Yes
Feature selection	No	Yes

⇒ ∴ if coefficient = 0 ⇒ Feature eliminated

⇒ Lasso = Shrinkage + Sparsity

↳ sparsity means many coeff. are zero

⇒ 4π Ridge, penalty : $\lambda \omega^2$.

∴ a coeff. of 10 contributes : 100λ [$\because \omega^2=100$]

while a " " 2 " " : 4λ [$\because \omega^2=4$]

∴ large coeff. are disproportionately penalized.

But, in Lasso, penalty : $\lambda \omega$

∴ $\omega=10 \Rightarrow 10\lambda$ } so, larger coeff. still incur a larger absolute penalty, but

$\omega=2 \Rightarrow 2\lambda$

unlike Ridge, the penalty doesn't grow quadratically.

⇒ Basically,

◦ Ridge: Keep all features, but reduce their influence.

◦ Lasso: Reduce their influence, and if a feature isn't useful enough, remove it completely.

⇒ What does sparsity mean? A coeff. vector is sparse when many of its values are exactly zero.

⇒ Why Lasso creates sparsity?

∴ $L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda |\mathbf{m}|$

⇒ $\frac{\partial L}{\partial m} = 0 \Rightarrow m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum (x_i - \bar{x})^2}$

∴ for $m > 0$:

$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum (x_i - \bar{x})^2}$

for $m = 0$:

$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$

for $m < 0$:

$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) + \lambda}{\sum (x_i - \bar{x})^2}$

So, basically, L1 geometry makes exact zero value possible for Lasso Regression.

Elastic Net Regression

⇒ Why do we need Elastic Net if we already had Ridge & Lasso? Ridge was not performing feature selection while Lasso was having issue to work with highly correlated features.

⇒ What is Elastic Net Regression? It's a regularized linear regression technique that combines L1 and L2 regularization, allowing it to simultaneously shrink coeffs, perform feature selection, and handle correlated features better than Lasso alone.

⇒ ElasticNet = L1 Regularization

+

L2 Regularization

$$\Rightarrow L = \sum_i (y_i - \hat{y}_i)^2 + a \sum_j \omega_j^2 + b \sum_j \omega_j$$

L2

L1

⇒ a = strength of L2 Regularization

b = " " L1

⇒ Also,

$$\lambda = a + b$$

$$\text{L1-ratio} = \frac{a}{a+b} = \gamma$$

$$\text{L1 contribution} = \lambda \gamma$$

$$\text{L2} = \lambda(1-\gamma)$$

⇒ what does L1-ratio = 0.5 means? The model gives equal weight to the two regularization penalties.

⇒ when to use ElasticNet? Highly correlated features.

Feature selection	Yes	No
Constant coeff.	Yes	No
Effect	Shrink coeff.	Shrink
Penalty	ω_2	ω_1
Loss	Loss	Loss

⇒ L1 coefficient = 0 = Feature eliminated

⇒ L2 = Shrinkage + stability

⇒ the ridge penalty: λ

⇒ L1 coefficient = 0 = Feature eliminated

⇒ L2 = Shrinkage + stability

⇒ L1 coefficient = 0 = Feature eliminated

⇒ L2 = Shrinkage + stability

⇒ L1 coefficient = 0 = Feature eliminated

⇒ L2 = Shrinkage + stability

⇒ L1 coefficient = 0 = Feature eliminated